STA2002 Probability and Statistics II
Assignment 1
Due date: **September 30 2023 11:59pm**

Please work on the assignment by yourself only and submit it on time. We follow a strict rule on academic honesty.

1. Let $X_1, X_2, \ldots, X_n$ be independent chi-square distribution with degrees of freedom $r_1, r_2, \ldots, r_n$, that is, $X_i \sim \chi^2(r_i)$, where $r_i > 0$, for $i = 1, \ldots, n$. Identify the distribution of $\sum_{i=1}^n X_i$ using the moment generating function technique.

   Since X_i are independent, for $t < 1/2$, we have

$$\begin{aligned}\mathbb{E}(e^{t\sum_{i=1}^n X_i}) &= \prod_{i=1}^n \mathbb{E}(e^{tX_i}) \\ &= \prod_{i=1}^n \frac{1}{(1-2t)^{r_i/2}} \\ &= \frac{1}{(1-2t)^{(\sum_{i=1}^n r_i)/2}},\end{aligned}$$

   which is the moment generating function of a chi-square distribution with $\sum_{i=1}^n r_i$ degrees of freedom. Since the moment generating function uniquely identifies the distribution, we conclude that the distribution of $\sum_{i=1}^n X_i$ is chi-square with $\sum_{i=1}^n r_i$ degrees of freedom.

2. The number of floods that occur in a certain region over a given year is a random variable having a Poisson distribution with mean 1.8, independently from one year to the other. Moreover, the time period (in days) during which the ground is flooded, at the time of an arbitrary flood, is an exponential random variable with mean 3. We assume that the durations of the floods are independent. Using the central limit theorem, calculate (approximately)

   (a) the probability that over the course of the next 20 years, there will be at least 19 floods in this region. Assume that we do not need to apply half-unit correction for this question.

   (b) the probability that the total time during which the ground will be flooded over the course of the next 120 floods will be smaller than 365 days.

   (a) Let X_i be the number of floods during the ith year. Then, the X_is are independent Poisson random variables with mean 1.8 and hence $E(X_i) = \mathrm{Var}(X_i) = 1.8$ for all i. We seek

$$\begin{aligned}\mathbb{P}\left(\sum_{i=1}^{20} X_i \geq 19\right) &\overset{CLT}{\simeq} \mathbb{P}(\mathrm{N}(20(1.8), 20(1.8)) \geq 19) \\ &= \mathbb{P}\left(\mathrm{N}(0,1) \geq \frac{19-30}{\sqrt{36}}\right) \\ &\simeq 1 - \Phi(-1.83) = \Phi(1.83) \overset{\text{Stand. Norm. Table}}{\simeq} 0.9664\end{aligned}$$

(b) Let Y_i be the duration of the ith flood. Then the Y_is are independent exponential random variables with mean 3. Because $\mathbb{E}(Y_i) = 3$ and $\mathrm{Var}(Y_i) = 3^2 = 9$, we calculate

$$\mathbb{P}\left(\sum_{i=1}^{120} Y_i < 365\right) \overset{CLT}{\simeq} \mathbb{P}(\mathrm{N}(120(3), 120(9)) < 365)$$
$$= \mathbb{P}\left(\mathrm{N}(0,1) < \frac{365 - 360}{\sqrt{120(9)}}\right) \overset{\text{Stand. Norm. Table}}{\simeq} 0.5517.$$

3. Suppose $X_1, X_2, \ldots, X_n$ are i.i.d. exponential random variables with pdf given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0\,, \\ 0 & \text{otherwise.} \end{cases}$$

We are interested in estimating the parameter λ using a number of methods.

(a) What is the method of moment estimator $\hat{\lambda}_{mom}$ of λ?
We set the sample mean to be equal to the true mean. That is,

$$\frac{1}{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n$$

and so $\hat{\lambda}_{mom} = \frac{1}{\bar{X}_n}$.

(b) What is the maximum likelihood estimator $\hat{\lambda}_{mle}$ of λ?
We first write out the likelihood function $L(\lambda)$. That is,

$$L(\lambda) = \prod_{i=1}^{n} f_X(X_i) = \lambda^n e^{-\lambda \sum_{i=1}^n X_i}$$

The log likelihood is given by $l(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n X_i$. We set the first derivative of l to be zero to find out the maximum, that is, we set

$$l'(\lambda) = 0$$
$$\frac{n}{\lambda} - \sum_{i=1}^{n} X_i = 0$$
$$\hat{\lambda}_{mle} = \frac{1}{\bar{X}_n}\,.$$

(c) Suppose the following $n = 6$ samples was generated from an exponential distribution with parameter λ as described in this question:

4.80 3.25 1.21 0.64 2.51 4.39

Compute $\hat{\lambda}_{mom}$ and $\hat{\lambda}_{mle}$. $n = 6$, $\bar{X}_n = 2.8$, so

$$\hat{\lambda}_{mom} = \hat{\lambda}_{mle} = \frac{1}{2.8} = 0.3571.$$

(d) Suppose that $n > 1$. Is $\hat{\lambda}_{mle}$ an unbiased estimator of λ? Why?
$\hat{\lambda}_{mle}$ is biased estimator of λ. Since $X_i \sim \mathrm{Gamma}(1, 1/\lambda)$, $\sum_{i=1}^n X_i \sim \mathrm{Gamma}(n, 1/\lambda)$. As a result,

$$\mathbb{E}(\hat{\lambda}_{mle}) = n\mathbb{E}\left(\frac{1}{\sum_{i=1}^n X_i}\right)$$
$$= n\int_0^\infty \frac{\lambda^n}{\Gamma(n)} x^{n-1-1} e^{-\lambda x}\, dx$$
$$= \frac{n\lambda}{n-1}\int_0^\infty \frac{\lambda^{n-1}}{\Gamma(n-1)} x^{n-1-1} e^{-\lambda x}\, dx$$
$$= \frac{n\lambda}{n-1} \neq \lambda.$$

4. Let $X_1, \ldots, X_n$ be an i.i.d. random sample from the uniform distribution on the interval (a, b). Find the method of moments estimators of a and b.

We have

$$\mathbb{E}(X) = \frac{1}{b-a}\int_a^b x dx = \frac{a+b}{2}$$
$$\mathbb{E}(X^2) = \frac{1}{b-a}\int_a^b x^2 dx = \frac{b^2+ab+a^2}{3}.$$

We solve the following equations for a, b

$$\hat{m}_1 = \frac{a+b}{2}, \quad \hat{m}_2 = \frac{b^2+ab+a^2}{3} \tag{1}$$

where $\hat{m}_1 = \frac{1}{n}\sum_i X_i$, and $\hat{m}_2 = \frac{1}{n}\sum_i X_i^2$.
From Equation (1), we get $a = 2\hat{m}_1 - b$. Plugging this into the equation of $\hat{m}_2$, we get

$$\begin{aligned} 3\hat{m}_2 &= b^2 + b(2\hat{m}_1 - b) + (2\hat{m}_1 - b)^2 \\ &= b^2 - 2\hat{m}_1 b + 4\hat{m}_1^2. \end{aligned}$$

Thus we have $b^2 - 2\hat{m}_1 b + 4\hat{m}_1^2 - 3\hat{m}_2 = 0$. By quadratic formula, we have $b = \hat{m}_1 \pm \sqrt{3(\hat{m}_2 - \hat{m}_1^2)}$. Then $a = 2\hat{m}_1 - b = \hat{m}_1 \mp \sqrt{3(\hat{m}_2 - \hat{m}_1^2)}$.

Since $a < b$, we have

$$\hat{a} = \hat{m}_1 - \sqrt{3(\hat{m}_2 - \hat{m}_1^2)}$$
$$\hat{b} = \hat{m}_1 + \sqrt{3(\hat{m}_2 - \hat{m}_1^2)}.$$

5. Find the maximum likelihood estimator of the unknown parameter θ where $X_1, X_2, \ldots, X_n$ is a sample from the distribution whose density function is

$$f_X(x) = \begin{cases} e^{-(x-\theta)} & \text{if } x \geq \theta \\ 0 & \text{otherwise.} \end{cases}$$

$$L(\theta) = \prod_{i=1}^n f_X(x_i) = \prod_{i=1}^n e^{-(x_i-\theta)} \mathbf{1}_{\{x_i \geq \theta\}}$$
$$\begin{aligned} l(\theta) = \log L(\theta) &= \log[\prod_{i=1}^n e^{-(x_i-\theta)} \mathbf{1}_{\{x_i \geq \theta\}}] \\ &= -\sum_i x_i + n\theta + \log \prod_{i=1}^n \mathbf{1}_{\{x_i \geq \theta\}} \\ &= -\sum_i x_i + n\theta + \log \mathbf{1}_{\{\min(x_i) \geq \theta\}} \end{aligned}$$

Since $\log \mathbf{1}_{\{\min(x_i) \geq \theta\}} = 0$, if $\min(x_i) \geq \theta$, and equals to $-\infty$ otherwise, and since $n\theta$ is increasing in θ, to maximize $l(\theta)$, we should choose $\hat{\theta}_{\text{mle}} = \min_i(X_i)$.

6. Let $X_1, \ldots, X_n$ be i.i.d. random samples from Uniform$(0, \theta)$. We saw in class that the mle $\hat{\theta}$ of θ is $\hat{\theta} = \max\{X_1, \ldots, X_n\}$. Show that this is a biased estimator of θ. Use the result to propose an unbiased estimator of θ. Hint: First find the cdf and then the pdf of $\max\{X_1, \ldots, X_n\}$. Then use it to calculate its mean.

We write $X_{(n)} = \max\{X_1, \ldots, X_n\}$. For $x \in (0, \theta)$, we have

$$\begin{aligned}\mathbb{P}(X_{(n)} < x) &= \mathbb{P}(X_1 < x, X_2 < x, \ldots, X_n < x) \\ &= \prod_{i=1}^{n} \mathbb{P}(X_i < x) \\ &= \left(\frac{x}{\theta}\right)^n.\end{aligned}$$

Therefore, the pdf of $X_{(n)}$ is

$$f_{X_{(n)}}(x) = \frac{nx^{n-1}}{\theta}, \quad 0 < x < \theta.$$

Its mean is

$$\mathbb{E}(X_{(n)}) = \int_0^\theta \frac{n}{\theta} x^n \, dx = \frac{n}{n+1}\theta \neq \theta,$$

and so $X_{(n)}$ is a biased estimator of θ. Since the estimator

$$\mathbb{E}\left(\frac{n+1}{n} X_{(n)}\right) = \theta,$$

$\frac{n+1}{n} X_{(n)}$ is an unbiased estimator of θ.

7. (Gaussian Mixture Model (GMM)) This question is about (a simplified version of) the Gaussian Mixture Model (GMM), which is a popular model in statistics, data science and machine learning. For example, it is used in image processing and various clustering algorithms. Suppose that K is a discrete random variable that can either be 0 or 1 with probability π_0 and π_1 respectively, that is,

$$K = \begin{cases} 0, & \text{with probability } \pi_0 = P(K = 0), \\ 1, & \text{with probability } \pi_1 = P(K = 1) = 1 - \pi_0. \end{cases}$$

Conditional on $K = k$ with $k \in \{0, 1\}$, the distribution of X is $N(\mu_k, \sigma_k^2)$, a normal distribution with mean μ_k and variance σ_k^2. That is,

$$\begin{aligned} X|K = 0 &\sim N(\mu_0, \sigma_0^2), \\ X|K = 1 &\sim N(\mu_1, \sigma_1^2). \end{aligned}$$

(a) Derive the joint density of (X, K). State clearly the support of (X, K) in the joint density. Hint: consider conditional distribution and the law of total probability.

Denote the joint density of (X, K) by $f_{(X,K)}(x, k)$. Also, denote the conditional density of $X|K = k$ by $f_{X|K}(x|k)$ for $k \in \{0, 1\}$. Clearly, k can only take values on $\{0, 1\}$. By the law of total probability,

$$\begin{aligned} f_{(X,K)}(x, k) &= f_{X|K}(x|k)\mathbb{P}(K = k) \\ &= \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(x-\mu_k)^2}{2\sigma_k^2}} \pi_k \\ &= \begin{cases} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{(x-\mu_0)^2}{2\sigma_0^2}} \pi_0, & \text{if } x \in \mathbb{R}, k = 0, \\ \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} \pi_1, & \text{if } x \in \mathbb{R}, k = 1, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

(b) Denote the distribution of $(X, K) \sim \text{GMM}(\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2)$. Note that π_1 can be omitted as a parameter since $\pi_1 = 1 - \pi_0$. Suppose that we have an i.i.d. random sample of size n of these n pairs $(X_1, K_1), (X_2, K_2), \ldots, (X_n, K_n)$. Each X_i belongs to either group 0 or group 1 depending on K_i. Using part (a), derive the maximum likelihood estimator for all the five parameters $\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2$. Hint: Let $n_0 = \sum_{i=1}^n \mathbf{1}_{\{K_i=0\}}$ and $n_1 = \sum_{i=1}^n \mathbf{1}_{\{K_i=1\}}$ be the number of X_i that

belongs to group 0 and group 1 respectively. You may find expressing the likelihood function in terms of n_0 and n_1 useful.

Let $n_0 = \sum_{i=1}^n \mathbf{1}_{\{K_i=0\}}$ and $n_1 = \sum_{i=1}^n \mathbf{1}_{\{K_i=1\}}$ be the number of X_i that belongs to group 0 and group 1 respectively. Then $n = n_0 + n_1$. Also, denote I_0 and I_1 to be respectively

$$I_0 = \{i;\ K_i = 0\}, \quad I_1 = \{i;\ K_i = 1\}.$$

The likelihood function is

$$\begin{aligned} L(\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2) &= \prod_{i=1}^n f_{(X,K)}(x_i, k_i) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma_0^2}}\right)^{n_0} \exp\left\{-\sum_{i\in I_0} \frac{(x_i-\mu_0)^2}{2\sigma_0^2}\right\} \pi_0^{n_0} \times \\ &\quad \left(\frac{1}{\sqrt{2\pi\sigma_1^2}}\right)^{n_1} \exp\left\{-\sum_{i\in I_1} \frac{(x_i-\mu_1)^2}{2\sigma_1^2}\right\} (1-\pi_0)^{n_1}. \end{aligned}$$

The log-likelihood function is

$$\begin{aligned} l(\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2) &= n_0 \log \pi_0 + n_1 \log(1-\pi_0) - \sum_{i\in I_0} \frac{(x_i-\mu_0)^2}{2\sigma_0^2} - \sum_{i\in I_1} \frac{(x_i-\mu_1)^2}{2\sigma_1^2} - (n_0/2)\log\sigma_0^2 \\ &\quad - (n_1/2)\log\sigma_1^2 - (n/2)\log(2\pi). \end{aligned}$$

We now take the partial derivative of l with respect to $\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2$. Note that for μ_0, σ_0^2 their partial derivatives in the log-likelihood function are exactly the same as covered in class (see LN4 note) or in the book Example 6.4-3 as if there are only n_0 samples. For μ_1, σ_1^2 their partial derivatives in the log-likelihood function is the same as covered in class (see LN4 note) or in the book Example 6.4-3 as if there are only n_1 samples. Therefore, the maximum likelihood estimators are

$$\begin{aligned} \hat{\pi}_0 &= \frac{n_0}{n}, \\ \hat{\mu}_0 &= \frac{1}{n_0}\sum_{i\in I_0} x_i, \\ \hat{\sigma}_0^2 &= \frac{1}{n_0}\sum_{i\in I_0} (x_i - \hat{\mu}_0)^2, \\ \hat{\mu}_1 &= \frac{1}{n_1}\sum_{i\in I_1} x_i, \\ \hat{\sigma}_1^2 &= \frac{1}{n_1}\sum_{i\in I_1} (x_i - \hat{\mu}_1)^2. \end{aligned}$$

(c) On Blackboard, there is a dataset called "GMM.csv", which is a comma separated file. This dataset contains $n = 100,000$ rows and 2 columns. The first column is the realizations of K, and the second column is the realizations of X. Using any computing language of your choice, e.g. Python/R/Julia/MATLAB/Excel, compute the maximum likelihood estimators derived in part

(b) using this dataset. Attach your code at the end of your assignment.

$$\hat{\pi}_0 = \frac{n_0}{n} = 0.78006,$$
$$\hat{\mu}_0 = \frac{1}{n_0} \sum_{i \in I_0} x_i = 19.97787,$$
$$\hat{\sigma}_0^2 = \frac{1}{n_0} \sum_{i \in I_0} (x_i - \hat{\mu}_0)^2 = 100.5967,$$
$$\hat{\mu}_1 = \frac{1}{n_1} \sum_{i \in I_1} x_i = 39.98855$$
$$\hat{\sigma}_1^2 = \frac{1}{n_1} \sum_{i \in I_1} (x_i - \hat{\mu}_1)^2 = 25.27242.$$